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Characterizing pigments, dyes, and fibers in murals from Bezeklik Cave 51 and Tuyuq Cave 2

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Turpan's mural paintings have been extensively studied, but inconsistent methodologies hinder comparison. We proposed a standardized framework by analyzing murals in Bezeklik Cave 51 and Tuyuq Cave 2, representing distinct yet interconnected styles and traditions in Turpan of Xinjiang, China. AMS ¹⁴C dating dates Cave 51 to 1025–1159 AD and Cave 2 to 424–564 AD (95.4% probability). Pigments from Cave 51 include gypsum, carbon black, atacamite, and red lead, with lead dioxide as a discoloration product. Cave 2 features gypsum, atacamite, hematite, and indigo. Both caves share a ground layer manufacturing technique: a coarse mud layer with wheat husks and straws, topped by a fine mud layer with hemp fibers, consistent with 4th–14th-century practices in northwestern China. Our results reveal spatial and temporal variations in material selection and technological practices, enabling direct comparisons and a better understanding of Silk Road cultural, artistic, and technological traditions.

Located in eastern Xinjiang, Turpan was an important transit point along the Silk Road, facilitating the exchange of cultures, materials, and technologies between the East and West. It has been home to diverse ethnic groups, especially since the Han Dynasty. The region's extreme aridity—with annual precipitation around 16.4 mm and evaporation exceeding 3000 mm, along with the annual average sunshine of 3360 h (the extreme temperature reached 47.8 °C in summer)¹—has contributed to the preservation of many human-modified landscapes, including ancient settlements, grottoes, and organic materials such as textiles, papers and documents, and food remains. Buddhism dominated Turpan until the mid-15th century when it was replaced by Islam. Numerous Buddhist grottoes and temples were constructed in Turpan, featuring intricately carved Buddhist sculptures and mural paintings. Although preservation varies, Turpan boasts a rich array of Buddhist remains, surpassing other regions in Xinjiang in both quantity and quality. However, only four grottoes from Turpan—Bezeklik, Tuyuq, Shengjinkou, and Bethhiah (Besh-ghar)—have been scientifically excavated and studied, with Bezeklik and Tuyuq particularly noted for their caves and impressive relics (see geographic locations of Bezeklik and Tuyuq grottoes in Fig. 1).

Previous scientific investigations of mural paintings in the Turpan Basin have been hindered by inconsistent analytical methods, limiting the ability to draw universally comparable insights into raw material selection and technological choices (see a summary of major findings in

Supplementary Tables 1–3). To address this issue, Wuhan University's Archaeology Department and the Academia Turfanica initiated a research project in July 2023. It aims to establish a standardized methodological framework to better understand inter- and intra-regional similarities and differences in mural painting preparation methods and techniques across the Turpan Basin. As a first step, we selected Bezeklik Cave 51 and Tuyuq Cave 2 for a comparative study. Although both caves have been analyzed previously, uncertainties persist regarding their dating and material characterization (such as pigment and dye identification and fiber analysis), highlighting the need for further investigation. We employed accelerator mass spectrometry carbon-14 (AMS ¹⁴C) dating, Raman spectrometry, high-resolution mass spectrometry (HRMS), Herzberg staining, and phytolith analysis to determine the mural painting's date and to characterize their pigments, dyes, and fibers. Although not the first multi-analytical examination of the mural paintings of the Turpan Basin, our study offers new insights and establishes solid, comparable findings on material selection and technological choices.

Bezeklik grottoes are situated at the cliff on the west bank of Mutougou in the Flaming Mountain (*huo yan shan*) of Turpan Basin; by contrast, Tuyuq grottoes are located on the cliffs on the east and west banks of the Tuyuq valley. The two grottoes are about 18 km apart in the straight-line distance. Both were severely damaged by the end of the 19th century AD, when many beautiful mural paintings and clay statues were destroyed or

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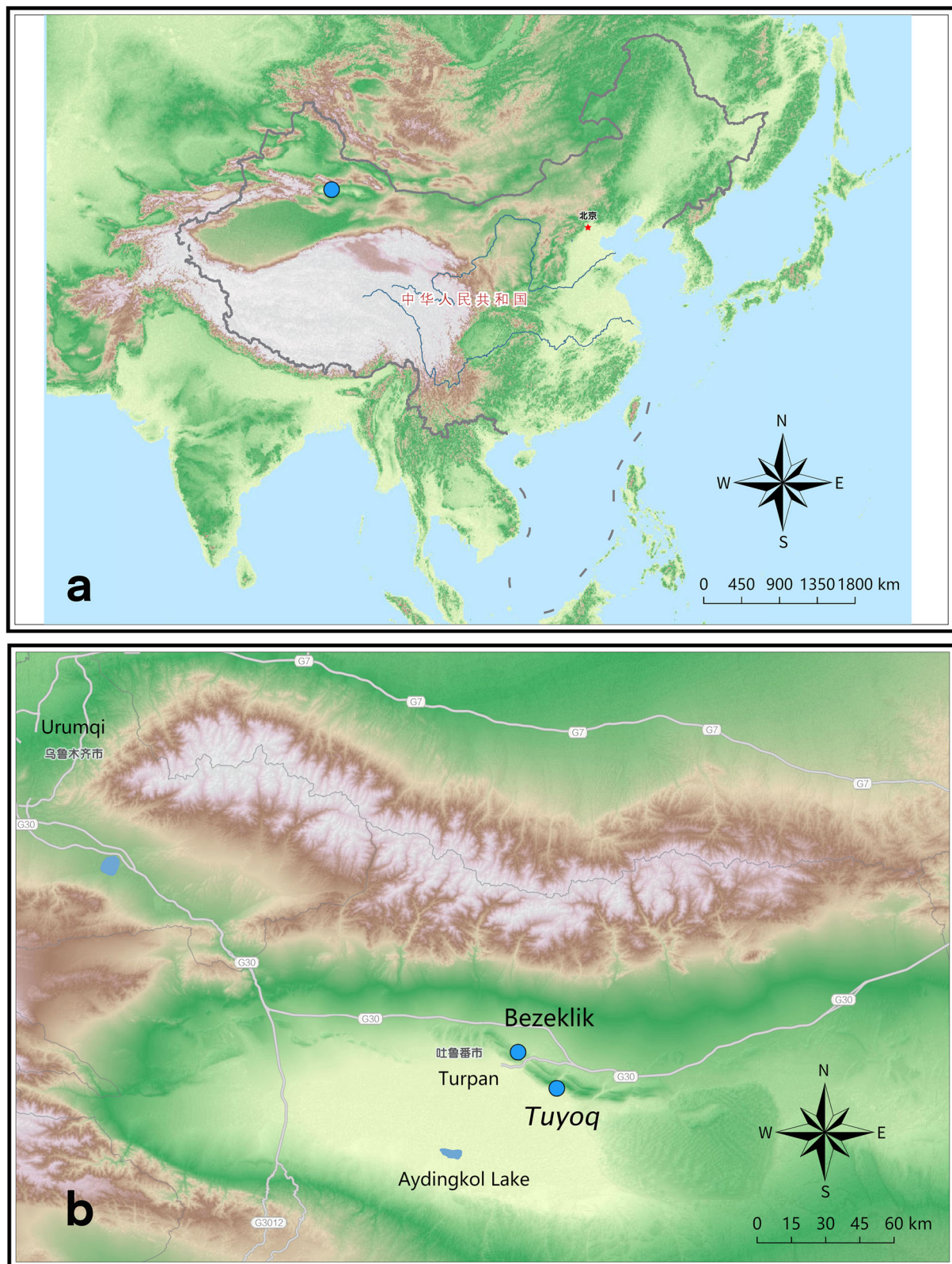


Fig. 1 | Maps showing geographic locations of the Bezeklik and Tuyuq Caves. **a** Their locations in the People's Republic of China. **b** Their locations in the Turpan Basin of Xinjiang.

looted. Salvage excavations were not conducted at Bezeklik grottoes until the People's Republic of China was established. In 1980 and 1981, archaeologists excavated caves and stupas at Bezeklik, yielding many documents, mural paintings, clay statues, and wooden architectural components². From 2010 to 2016, a joint team of the Research Center for Frontier and Ethnic

Archaeology from the Chinese Academy of Social Sciences and the Academia Turfanica carried out excavations in the Tuyuq grottoes³. Bezeklik and Tuyuq grottoes are important windows to probe into the changing religious and political powers that controlled the Turpan Basin in historical periods⁴. Furthermore, the artistic style of images in murals and the use of

pigments/dyes reflect the exchange of cultures and materials along the Silk Road. For example, mineral pigments (such as lapis lazuli and atacamite) and gold foils are believed to have indicated the material exchanges between Turpan and the Central Plains, Afghanistan, and India^{5,6}.

Studies on Bezeklik and Tuyuq grottoes can be subdivided by their objects, goals, and/or methodologies: (1) periodization, layout, and function of grottoes^{4,7}; (2) dating and interpretation of wooden block and paper prints unearthed from the grottoes^{8–10}; (3) preparation methods of mural paintings^{5,6,11–13}; and (4) conservation and restoration of grottoes and mural paintings^{14,15}. Our research focuses on the third category, specifically on the investigations of pigments, dyes, fibers, and relevant issues involving scientific methods and techniques. Most scientific investigations of Bezeklik and Tuyuq grottoes reported identification results of pigments and dyes from mural paintings. Below is a concise summary of their findings (see details of identified pigments and dyes, as well as the methods and techniques employed, in Supplementary Tables 1–3).

- Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) and anhydrite (CaSO_4) were often identified on the mural paintings as the white pigment; in contrast, kaolin ($2\text{SiO}_2 \cdot \text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$), calcite (CaCO_3), lead white [$2\text{PbCO}_3 \cdot \text{Pb}(\text{OH})_2$], or anglesite (PbSO_4) was used to paint white only occasionally^{6,11–13,16}.
- The black pigment is mainly carbon black (C)^{5,12,13}. Sometimes, discoloration products of lead oxides are identified as black or dark-colored pigments. For example, lead sulfide (PbS) and lead dioxide (PbO_2), which are respectively discoloration products of lead white and red lead (Pb_3O_4), have been confirmed as the black or brownish-black color^{6,11}.
- Red lead (Pb_3O_4), hematite (also known as red ochre, iron red or iron oxide, $\alpha\text{-Fe}_2\text{O}_3$), and cinnabar (HgS) were used to paint red at Tuyuq^{5,14}. By contrast, only red lead and hematite have been identified at Bezeklik^{11–13,16}. Besides, red pigments is often mixed with gypsum, quartz (SiO_2), or carbon black to create different shades of red^{11–13}.
- The green pigment is mainly identified as atacamite [$\text{Cu}_2(\text{OH})_3\text{Cl}$]^{5,12–14,16}. Another common mineral green pigment—malachite [$\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$] is used less frequently¹¹.
- The blue pigment in Tuyuq grottoes is reportedly prepared from lapis lazuli [$(\text{Na,Ca})_8(\text{AlSiO}_4)_6(\text{SO}_4,\text{Cl,S})_2$]^{5,6,14}, while the blue color in Bezeklik grottoes is made from azurite [$\text{Cu}_3(\text{CO}_3)_2(\text{OH})_2$] and malachite^{6,12,13}. Some argued that Bezeklik Cave 15 and Cave 20 used lapis lazuli as blue pigments, but no solid evidence was offered³.
- The yellow pigment in Tuyuq Cave 2 is orpiment (As_2S_3)⁶, and the gold color is prepared from gold in both Bezeklik and Tuyuq grottoes^{5,6,14}.

In summary, the colors in the mural paintings of Bezeklik and Tuyuq grottoes were predominantly prepared from mineral pigments, some of which were used more consistently than others, while organic dyes were reported only occasionally (they were not reported for Bezeklik Cave 51 and Tuyuq Cave 2, the focus of this paper). We observed that the characterization of pigments and dyes varied in accuracy and reliability; sometimes, identification relied solely on a single technique (e.g., elemental compositional analysis; see Supplementary Tables 1–3). Although these characterizations are not necessarily incorrect, they indicate differing levels of confidence across case studies.

In addition to pigments and dyes, the ground layer of mural paintings has also been studied, though the reliability of these investigations varies due to differences in methodology, sample availability, or analytical techniques. The ground layer, also referred to as the mud or earthen plaster, is typically divided into two layers: the *cu ni* (coarse mud) layer and the *xi ni* (fine mud) layer, composed of different materials such as clay, sand, and fibers¹⁷. Notably, while some case studies provide detailed information on the structure of the ground layer and the types of clay and sand used, the plant materials incorporated into the ground layer are poorly understood^{15,11,13,17,18}. Only a few researchers have presented microscopic details of plant identification^{13,18}. Methods and techniques such as Herzberg staining and phytolith analysis, which are commonly used in the identification of ancient paper and non-wood fibers, are rarely employed in identifying fibers in the ground layer¹⁸.

Methods

Materials

Based on our literature review, we propose a comparative study using unified methods to analyze the similarities and differences in mural materials and preparation techniques between the Bezeklik and Tuyuq grottoes in the Turpan Basin. This study aims to establish a consistent methodological framework for future scientific investigations of ancient grottoes in the Turpan Basin. Building on our previously proposed framework for analyzing pigments, dyes, and fibers in ancient Chinese paper-based materials¹⁹, we argue that a similar approach is essential for comparative studies of Silk Road mural paintings. Such a framework will enable us to assess how mural painting traditions in the Turpan Basin align with those elsewhere along the Silk Road and explore the cultural and technological factors behind these practices. To address these questions, we selected mural painting fragments from Bezeklik Cave 51 and Tuyuq Cave 2, both of which have not been thoroughly studied in terms of their pigments, dyes, or ground layers. These sites provide an ideal opportunity to apply and refine our methodological framework, offering new insights into the artistic and technological traditions of the region.

Bezeklik Cave 51 was repurposed from a monk's dwelling into a place of worship, featuring a rectangular longitudinal vaulted structure with front and rear chambers². The entrance to the rear chamber is located at the center of the front chamber's main wall. The rear room is empty due to theft and has no remaining artifacts. There are no traces of mural paintings, but a rectangular recess on the back wall, likely to be a Buddha niche¹⁶. In the front chamber, images on the main and right walls are severely damaged. The signs of cutting still remain on the right wall, with the stolen mural paintings documented in "Wall Paintings from Ancient Shrines in Central Asia"¹⁶. The left wall paintings are relatively well-preserved. Scholars have identified the subject of these paintings as scenes from the Buddhist Lotus Sutra¹⁶. The ceiling murals in the cave represent the best-preserved section, depicting Thousand Buddhas with a well-organized layout, vivid colors, and clear, discernible imagery. Figure 2 shows the remaining mural paintings in Bezeklik Cave 51.

Tuyuq Cave 2 is a vaulted central pillar cave, with the Buddha seat still intact in the niche of the central pillar's main wall. Mural paintings in Cave 2 depict Buddhist themes in the style of Kucha⁷, in a poor state of preservation. Paintings depicting motifs of Thousand Buddhas remained only on the vaulted ceiling of the central pillar passageway and the side walls. A large portion of the mural paintings has severely peeled off, with the remaining parts suffering from oxidation. The Buddhas in the paintings all wear draped shoulder-style robes and are seated on lotus flowers. The robes are in six colors: orange, green, red, blue, ochre, and white, arranged diagonally in a cyclical pattern⁷. Figure 3 shows the remaining mural paintings in Tuyuq Cave 2.

The mural paintings in Bezeklik Cave 51 and Tuyuq Cave 2 are rich in content, diverse in color, and intricately detailed. The surviving fragments retain substantial research significance despite severe damage from natural or human causes. Figures 4 and 5 are over 20 pieces of mural paintings collected from Cave 51 and Cave 2, respectively, containing white, black, red, yellow, green, and blue (bluish-black) colors. Most samples contain at least two different types of plants or fibers (observed under a low-power microscope without damaging the samples), which are ideal for the radio-carbon dating test.

Accelerator mass spectrometry carbon-14 dating

Plant fibers were noticed in the ground layer when the fragments of mural paintings in Bezeklik Cave 51 and Tuyuq Cave 2 were examined under an optical microscope in the laboratory at Wuhan University's Archaeology Department. Fiber specimens were collected from each cave and sent to the Beta Analytics Laboratory for accelerator mass spectrometry carbon-14 dating. The standard pretreatment procedure for plant fibers involved an acid/base/acid treatment. Fibers were crushed into 1–2 mm particles and heated in a 0.1 N hydrochloric acid solution at 70 °C for 1–2 h. They were rinsed to neutrality, followed by multiple repetitions using a 1–2% base solution (50/50 wt sodium hydroxide) at 70 °C until no color change or reduction

Fig. 2 | Mural paintings in Bezeklik Cave 51. a Cave ceiling murals featuring pagodas amid Thousand-Buddha designs. **b** Details of pagoda. **c** Details of the meditating Buddha, its face deliberately vandalized. **d** Well-preserved Lotus Sūtra scenes on the antechamber's left wall feature curtain drapes and panoramic landscapes.



occurred. Rinsing to neutrality was repeated, followed by a final hot acid wash (0.1 N hydrochloric acid) to ensure all base solutions were neutralized, followed by rinsing until pH neutrality was restored. The samples were dried for 12–24 h and examined under a microscope for purity and integrity. Subsequently, the samples were combusted in 100% 9.9999 oxygen.

Raman spectroscopy

An XploRA PLUS laser confocal Raman spectrometer manufactured by HORIBA Jobin Yvon was used to characterize and identify the colors on mural paintings. Three excitation light wavelengths (532, 638, and 785 nm) were used to ensure the optimal Raman signals were obtained. The analysis was conducted with a spot size smaller than 2 μm , and the laser power that reached each sample was between 0.25 and 0.5 mW. Background spectra of water and carbon dioxide were collected at room temperature. The Raman spectra in this study were subjected to smoothing and baseline correction.

High-resolution mass spectrometry

A blue (sometimes bluish-black) color was noticed on mural paintings in Tuyuq Cave 2. Raman analysis of the blue or bluish-black color indicates

indigo, an organic dye, was used. High-resolution mass spectrometry (HRMS) was used to cross-check this statement. Using the Bruker Compact TOF high-resolution mass spectrometer, HRMS analysis was conducted in positive electrospray ionization mode (ESI+) with a scanning range of 50–1500 m/z. Samples of blue and bluish-black colors from the Tuyuq Cave 2 were taken using tweezers and immersed in HPLC-grade methanol in sample tubes. Extraction was performed with an ultrasonic program at 70 °C for 30 min, followed by filtration or centrifugation to obtain the samples. Identification was based on the main ionization products from the ESI mass spectra, utilizing the Bruker Data Analysis software for analysis.

Identification of the fibers pre-treated with Herzberg staining

Under microscopic observation of the sampled mural painting fragments, it was noted that the ground layer of both caves contained plant fibers. A Herzberg staining reagent was prepared to confirm the type of plant fibers present. The type of plant fibers was identified by observing the stained color and fiber morphology^{19–21}. The general formula and procedure for Herzberg staining reagent are as follows²²: (1) 20 g of anhydrous zinc chloride (ZnCl_2), 10 ml of distilled water, prepared to form a zinc chloride solution; (2) 2.1 g of

Fig. 3 | Remaining mural paintings in Tuyuq Cave 2. **a** Vaulted ceiling retains Thousand Buddha murals, each seated on lotus platforms. **b** Thousand Buddha motifs (seated in meditation) adorn the side walls. **c** A Buddha figure dressed in a bluish-black robe. **d** A Buddha figure dressed in a whitish robe.



potassium iodide (KI) and 0.1 g of iodine (I_2) in 5 ml of distilled water; (3) after mixing the two solutions evenly, allow them to stand in the dark for 12–24 h, then collect the supernatant, which constitutes the Herzberg staining reagent.

Extraction and identification of phytoliths

Plant identification is also made by looking for characteristic phytoliths that can only be produced by specific species or families²³. The plant tissues from two caves were treated with the wet oxidation method by immersing them in 65% concentrated nitric acid (HNO_3) with water-bath heating at 80 °C for 5 h, while reference material (modern wheat specimens) were treated with the dry ashing method, ashed in the muffle furnace for 5 h at 500 °C^{24,25}. We prepared microscope slides to observe and document phytoliths.

Results

AMS ^{14}C dates

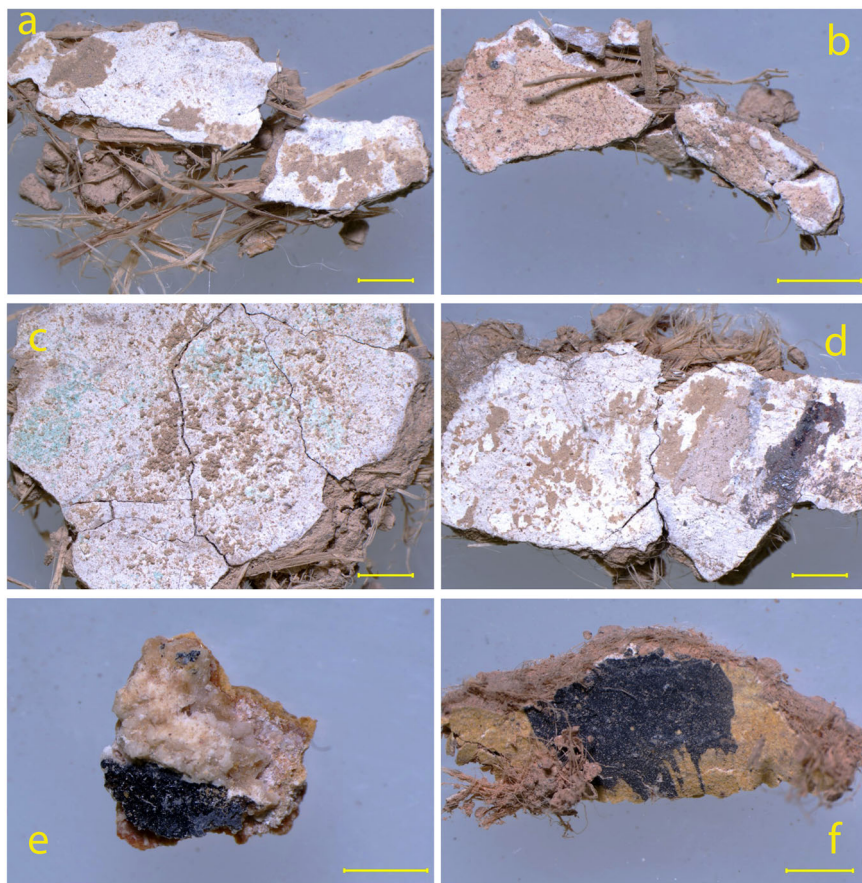
The results of accelerator mass spectrometry carbon-14 dating are shown in Fig. 6. In ancient China, plant fibers were often mixed with mud to prepare the ground layer and mainly sourced from herbaceous plants such as straw,

grass, and hemp fibers. Thus, the AMS ^{14}C dates of plant fibers indicate the manufacturing dates of the mural paintings and provide the first absolute dates for the two caves. The plant fibers sampled from Bezeklik Cave 51 are radiocarbon dated to 960 ± 30 years before present, with a calibrated age range of 1025–1159 AD (95.4% probability ranges, Beta-681886), falling within the Uyghur Gaochang period (mid-9th to 13th century). The Bezeklik grottoes were initially built around 499–640 AD (the Qu Gaochang period), serving as royal temples, and continuing into the Uyghur Gaochang period². Judging from the AMS ^{14}C dates, the murals in Cave 51 were prepared in the middle Uyghur Gaochang period. The plant fibers sampled from Tuyuq Cave 2 date back to 1570 ± 30 years before present, with a calibrated age range of 424–564 AD (95.4% probability ranges, Beta-681887), corresponding to the periods of Gaochang Prefecture (327–442 AD), Northern Liang Gaochang (442–460 AD), and Gaochang Kingdom (460–640 AD)⁷. The Tuyuq grottoes were initially constructed around the 5th century AD⁵. Therefore, Cave 2 was among the early caves at Tuyuq.

Pigment/dye identification for Bezeklik Cave 51

The Raman spectra of the pigments and dyes on mural paintings in Bezeklik Cave 51 are shown in Fig. 7.

Fig. 4 | Fragments of mural paintings in Bezeklik Cave 51 (scale bar = 2 mm). a White and yellow pigments. **b** Yellow pigment. **c** White and green pigments. **d** White and black pigments. **e** Black pigment. **f** Black and yellow pigments.



The white. The Raman spectrum of the white color shows peaks at 420, 502, 610, 629, 677, 1018, 1129, and 1161 cm^{-1} , matching the characteristic peaks of anhydrite or anhydrous calcium sulfate (CaSO_4)²⁶. Anhydrite and gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) can be distinguished by the position of the strongest band attributed to the vibration $\nu_1(\text{a}_1)$ SO_4^{2-} : 1018 cm^{-1} for anhydrite while 1007 cm^{-1} for gypsum^{27,28}. Gypsum can convert to anhydrite through dehydration (losing its water content); conversely, anhydrite can convert back to gypsum through hydration when exposed to water²⁹. Considering the extreme aridity and high temperatures in Turpan, as well as the poor water solubility and bonding properties of anhydrite, we speculate that gypsum was initially applied as a white pigment. Over time, it converted into anhydrite by losing its water content. Based on our findings and literature review, we suggest that gypsum was used both as a white pigment and as a white plaster for the ground layer—a conclusion that also applies to Tuyuq Cave 2, which we will discuss later.

The yellow. The Raman spectrum of the yellow color shows peaks at 121, 151, 227, 314, 389, and 548 cm^{-1} , which are characteristic of red lead (or ‘minium’ in mineral form, Pb_3O_4)²⁶. The Raman peak at 121 cm^{-1} is caused by the bending vibrations of O–Pb–O, and the peak at 548 cm^{-1} is due to stretching vibrations of Pb–O bonds³⁰. Red lead is a mineral pigment commonly used for red or orange hues. It appears yellow in the mural paintings of Cave 51, possibly because red lead was oxidized to form lead (II) oxide (massicot, PbO)³¹.

The green. The Raman spectrum of the green color can be identified as atacamite [orthorhombic, $\text{Cu}_2(\text{OH})_3\text{Cl}$], which reportedly shows Raman peaks at 139, 149, 218, 266, 358, 411, 449, 512, 595, 820, 843, 911, 974, 3329, 3349, 3433 cm^{-1} ³⁰. Atacamite (orthorhombic), clinoatacamite (monoclinic), and botallackite (monoclinic) can be distinguishable from

one another by their Raman spectra in the 100–1000 cm^{-1} and 3100–3700 cm^{-1} ^{30,32,33}.

The black. The Raman spectrum of the black color shows two peaks at 1321 and 1583 cm^{-1} , matching the Raman peaks of carbon black (C) (1315 and 1600 cm^{-1})³⁴. The two broad peaks are attributed to the D-band and G-band of carbon black³⁵. In addition to carbon black, we identified lead dioxide (PbO_2) as black, showing Raman peaks at 169, 230, and 508 cm^{-1} . We suggest that lead dioxide is a discoloration product of red lead³⁶.

Pigment/dye identification for Tuyuq Cave 2

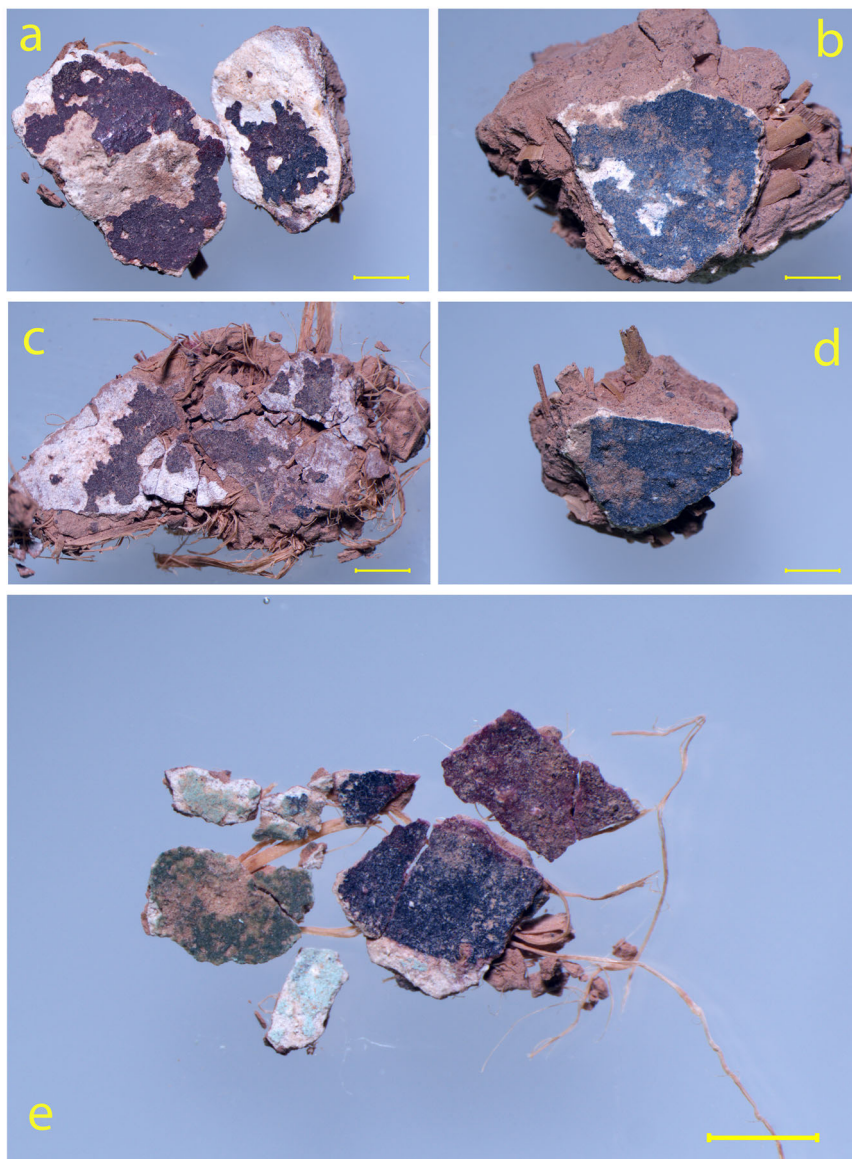
The Raman spectra of the pigments and dyes on mural paintings in Tuyuq Cave 2 are shown in Fig. 8.

The white. Raman spectrum of the white color shows peaks at 419, 501, 613, 629, 1018, 1130, and 1160 cm^{-1} , indicating the presence of anhydrite in the white paint²⁶. It was gypsum, we suggest, that was initially used to paint the white color.

The red. Raman spectrum of the red color shows peaks at 150, 294, 411, 612, 1281, and 1594 cm^{-1} , matching the characteristic peaks of hematite ($\alpha\text{-Fe}_2\text{O}_3$)³⁷. The most intense peaks of hematite are concentrated in the 200–650 cm^{-1} range, with strong signals at 294, 411, and 612 cm^{-1} . Another main peak near 225 cm^{-1} is not always observed or may appear with a medium to weak intensity in the spectrum³⁷.

The blue and bluish-black. We collected two ‘blue’ pigment samples, one is blue and the other is bluish-black. Raman spectrum of the bluish-black color shows peaks at 93, 135, 251, 547, 596, 676, 758, 1230, 1310,

Fig. 5 | Fragments of mural paintings in Tuyuq Cave 2 (scale bar = 2 mm). a Red pigment. **b** Blue pigment. **c** White and red pigments. **d** Bluish black pigment. **e** Green, bluish black and red pigments.



and 1571 cm^{-1} . The Raman spectrum of the blue color shows peaks at $547, 594, 1248, 1363, 1580$, and 1694 cm^{-1} . Despite the color differences, both colors show about the same Raman peaks, implying the same substance for the blue or bluish-black color. The substance matches the Raman peaks of indigo [$(\text{C}_{16}\text{H}_{10}\text{N}_2\text{O}_2)$], which are noticed at $98, 136, 172, 181, 236, 253, 265, 277, 311, 320, 468, 546, 599, 676, 758, 862, 871, 1015, 1149, 1226, 1248, 1310, 1363, 1461, 1572, 1584, 1626$, and 1701 cm^{-1} ²⁶. The range of $1554\text{--}1599\text{ cm}^{-1}$ is the most characteristic band of indigo. The most obvious Raman peak near 1571 cm^{-1} is attributed to the stretching vibrations of the conjugated system of C=C, C=O and N-H groups³⁸. Indigo differs significantly from mineral blue pigments such as lapis lazuli (with characteristics Raman peaks at $258, 548, 803, 1096$, and 1635 cm^{-1})^{39–41} and azurite [$2\text{CuCO}_3\cdot\text{Cu}(\text{OH})_2$], with characteristic Raman peaks at $145, 180, 250, 284, 335, 403, 545, 746, 767, 839, 940, 1098, 1432$, and 1573 cm^{-1}]⁴². We conclude that the blue and bluish-black colors were prepared from organic dye indigo instead of mineral pigments.

The green. Raman spectrum of the green color shows peaks at $510, 818, 911, 983, 2060, 3345$, and 3434 cm^{-1} , matching characteristic peaks of mineral atacamite⁴².

As for the white plaster of the ground layer in Cave 2, we suggested that it was plastered with gypsum, the same as in Cave 51.

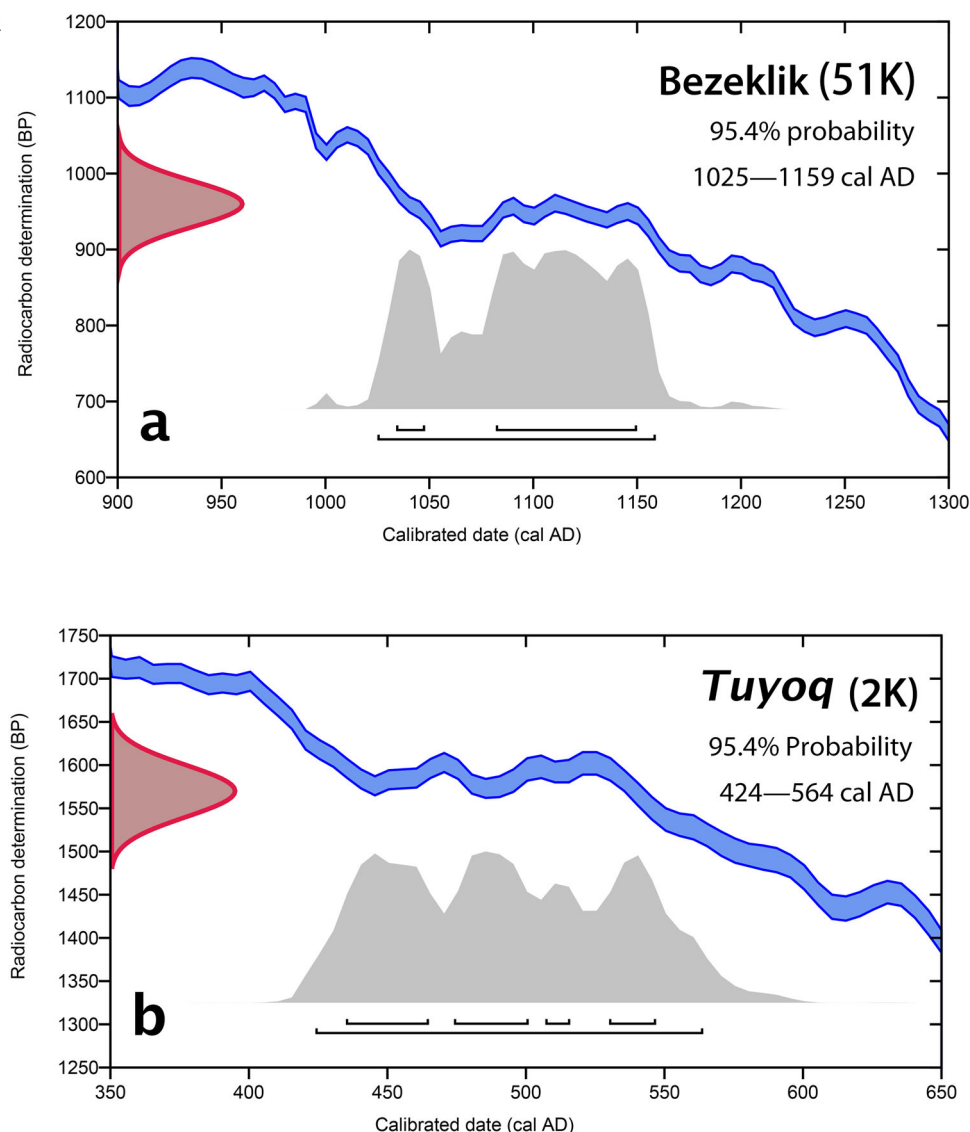
HRMS analysis

HRMS analysis was conducted to cross-check the identification results of Raman analysis of the blue and bluish-black colors in Tuyuq Cave 2. HRMS has been proven to be a powerful tool for identifying organic dyes with high accuracy^{19,43,44}. In Fig. 9, the molecular weights of the blue and bluish-black colors are 263.0808 and 263.0818 , respectively, with errors less than 0.001 compared to the standard indigo (263.0815)^{19,45}. We are confident that indigo was used to paint blue or bluish-black in Tuyuq Cave 2.

Analysis of fibers in the ground layer

Hertzberg staining was used as a quick, cost-effective, and minimally invasive analytical approach to help identify non-wood plant fibers and understand the ground layer preparing techniques^{20,46,47}. The stained plant fibers sampled from the ground layers of mural paintings in Bezeklik Cave 51 and Tuyuq Cave 2 show a wine-red to dark-red color (see Figs. 10 and 11), which differs significantly from (1) bamboo fibers showing a bright yellow color due to the high lignin content and (2) grass fibers showing a blue or bluish color due to moderate lignin content²¹. The stained color is consistent

Fig. 6 | AMS ^{14}C dates of the plant fibers sampled from mural painting fragments. a From Bezeklik Cave 51. b From Tuyuq Cave 2.



with bast fibers such as cotton or flax, both containing low lignin content^{20,46,48}.

Morphologically, the plant fibers in the ground layers of the two caves are cylindrical with uniform widths, distinct from cotton and ramie fibers^{49–52}. Furthermore, the transverse marks on the fiber surface are clearly noticed, while longitudinal marks are not. Cuticle is not noticed on the fiber's outer wall, and thin-walled cells or crystal particles are missing⁴⁸. We infer that the plant fibers were sourced from *ma*, a generic term in Chinese used to indicate hemp, ramie, and flax. We are inclined to believe that hemp fibers were used. Our statement is compatible with fiber morphology⁵³.

Two other lines of evidence are also in support of the use of hemp fibers: (1) technical examination of the ground layer in grottoes along the Silk Road¹⁷, and (2) the long-term cultivation and use of hemp (*Cannabis*) fibers in ancient Turpan^{54–56}. According to the most comprehensive analytical studies¹⁷, the ground layer in grottoes along the Silk Road is usually prepared in two main steps: first, mud mixed with wheat straws is plastered onto the wall to form a so-called *cu ni* (coarse mud) layer; what follows next is to apply mud mixed with fibers of *ma* onto the coarse mud layer to form the so-called *xi ni* (fine mud) layer. The Herzberg staining test results in our study are compatible with hemp fibers belonging to *xi ni* layer.

We also discovered husks and straws in the coarse mud layer of the ground layer of Cave 51 and Cave 2. Figure 12a, b shows husks likely from common wheat (*Triticum aestivum*) in the ground layer of Bezeklik Cave 51.

Phytolith analysis was conducted to confirm the inference. We extracted and identified articulated elongate dendritic phytoliths in husks sampled from Bezeklik Cave 51 (Fig. 12c, d) and Tuyuq Cave 2 (Fig. 12e, f)⁵⁷. A number of papillae are distributed between the dendritic phytoliths, and several pits can be observed around the rim of each papilla (Fig. 12d, f)^{58–60}. In particular, the undulating wave patterning (caused by the interlock of dendritic phytoliths) and the shape and ornamentation of papillae cells help distinguish wheat from other Pooideae grasses⁵⁸. Comparing them with phytoliths extracted from reference material (modern common wheat, see Fig. 12g, h), we can conclude the use of common wheat in the coarse mud layer of the ground layer of the two caves.

Our study suggests that wheat husks were mixed with mud to prepare the ground layer of mural paintings in Bezeklik and Tuyuq grottoes, just as those noticed in mural paintings of other grottoes^{5,11}. A previous study of adobe samples from Bezeklik grottoes reveals that grape (*Vitis vinifera*) and cereals, including broomcorn millet (*Panicum miliaceum*), common wheat (*Triticum aestivum*), and barley (*Hordeum vulgare* var. *coeleste*), were cultivated in the region during the 9th to 13th century AD⁶¹. Pollen of *Vitis* and leaf epidermis of *Triticum* were also found in Tuyuq grottoes, indicating the wheat and grape cultivation in 5th century AD in this region⁶². Our phytolith analysis is well-compatible with the findings from this study.

Dighe et al.¹⁸ once identified rice (*Oryza sativa*) straw, roots, and husks in the ground layer (earthen plaster) of mural paintings, originally painted in

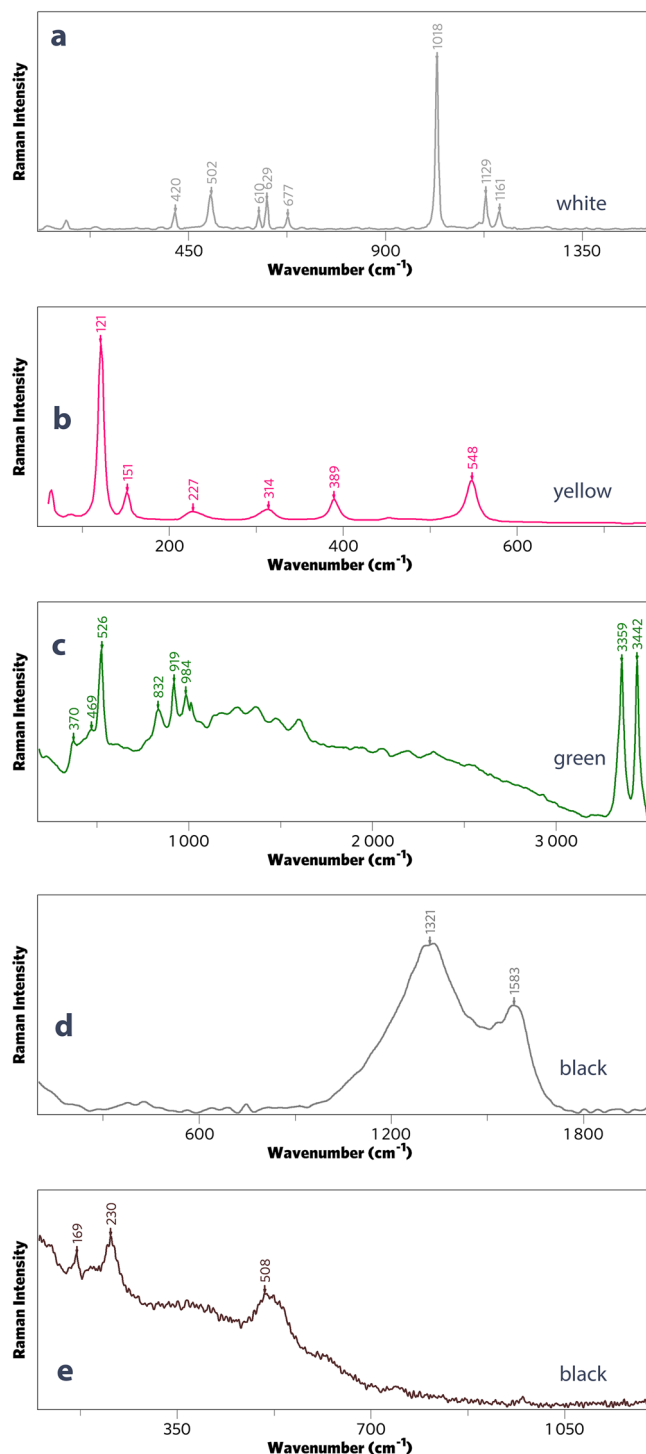


Fig. 7 | Raman identification of pigments on mural paintings in Bezeklik Cave 51. a Anhydrite as white. **b** Red lead as yellow. **c** Atacamite as green. **d** Carbon black as black. **e** Lead dioxide as black.

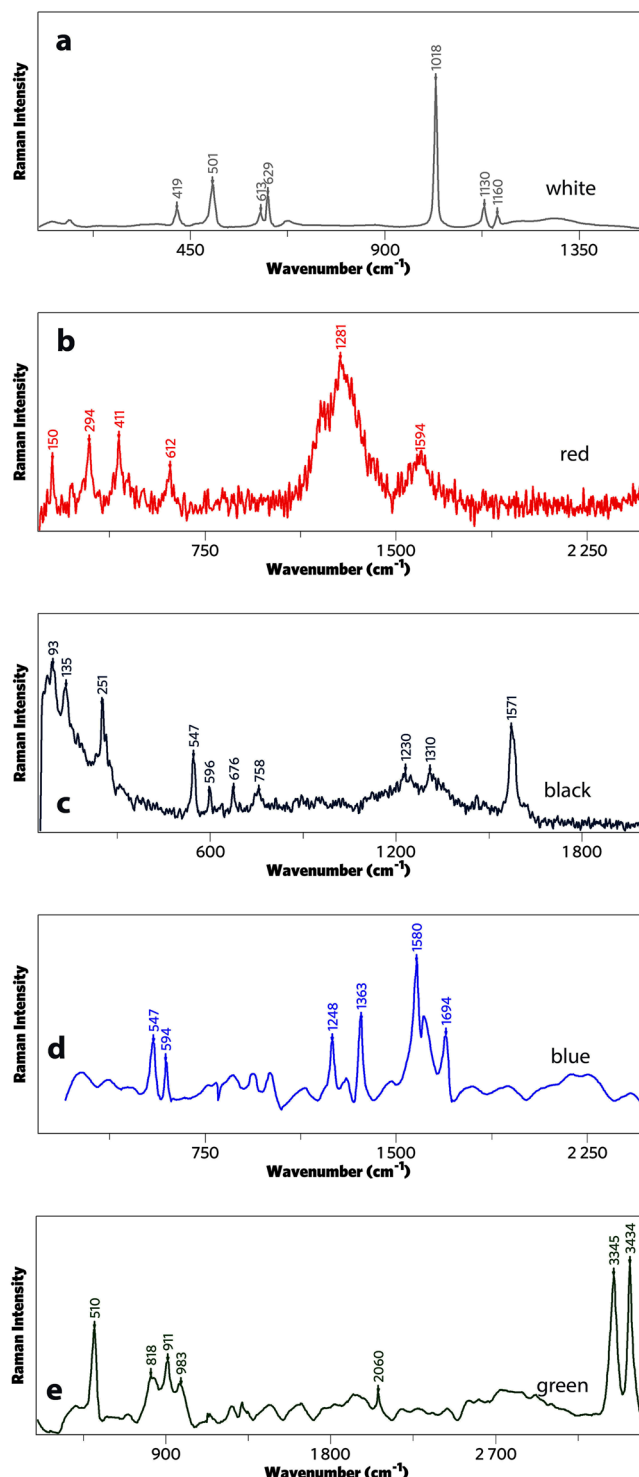


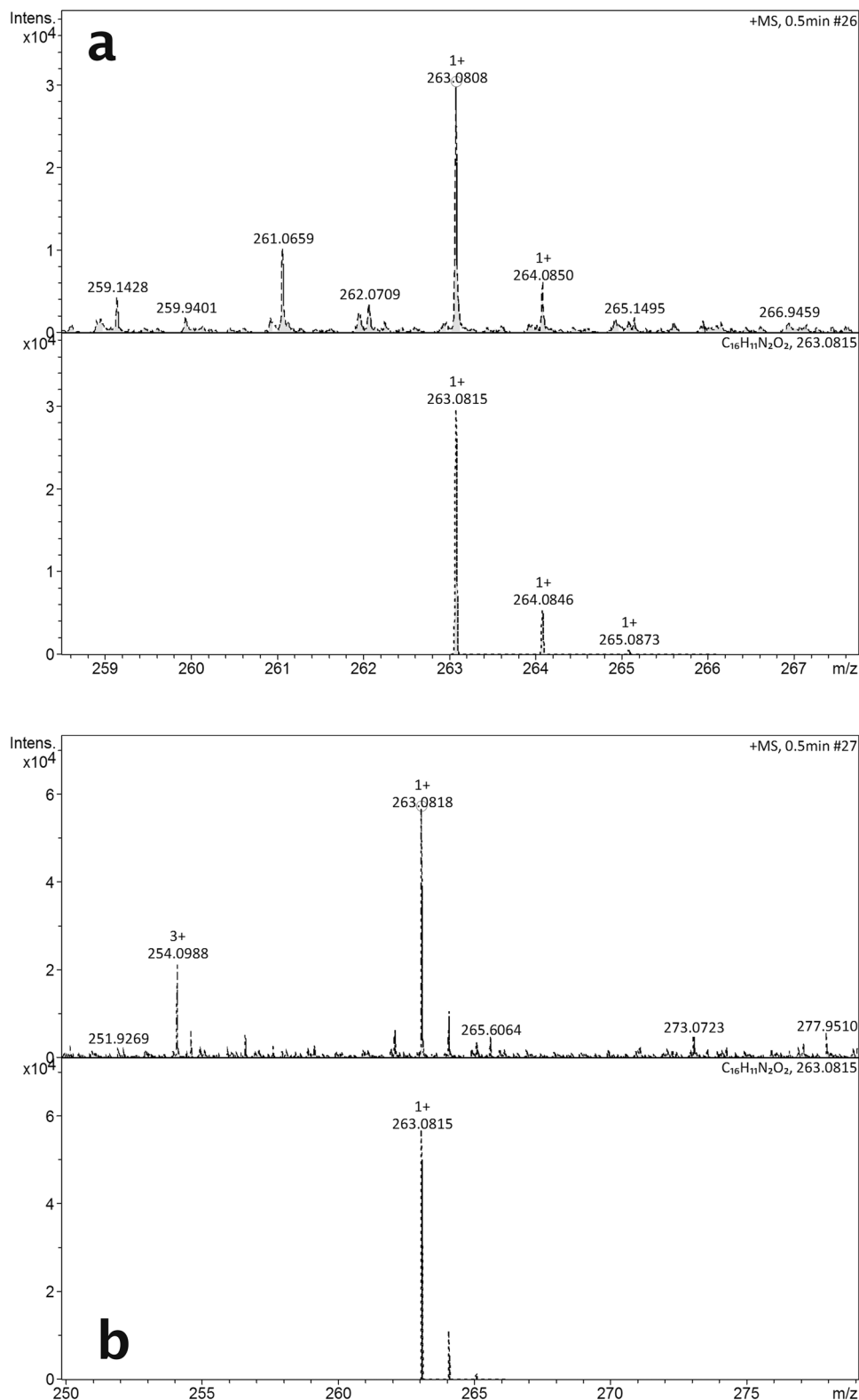
Fig. 8 | Raman identification of pigments on mural paintings in Tuyuq Cave 2. a Anhydrite as white. **b** Hematite as red. **c** Indigo as bluish black. **d** Indigo as blue. **e** Atacamite as green.

Cave 18 of Bezeklik grottoes, through phytolith analysis, Fourier transform infrared spectroscopy (FTIR), and scanning electron microscopy (SEM). Dighe et al.¹⁸ argue that rice was cultivated in the Turpan Basin and consumed as food or mixed with mud to prepare the ground layer of mural paintings. The use of rice by-products in the ground layer of mural paintings in Xinjiang has not been widely reported or confirmed. Our study here highlights the role of wheat by-products in preparing the ground layer of mural paintings.

Animal fibers in the ground layer

In addition to plant fibers, we noticed animal hairs in the ground layer of mural paintings in Bezeklik Cave 51, which show clear structures of hair scales and medullary cavities⁶³ (see Fig. 13). We believe that animal hairs were used, given the local practice of adding animal hairs to clay to prepare a mud paste. However, the specific animal species these hairs belong to requires further investigation. We have not yet identified animal fibers in the ground layer of mural paintings in Tuyuq Cave 2.

Fig. 9 | HRMS identification of the blue and bluish-black dyes on mural paintings in Tuyuq Cave 2, with indigo as reference material. a Blue dye. b Bluish dye.



Discussion

The identification results of pigments, dyes, and fibers for Bezeklik Cave 51 and Tuyuq Cave 2 are summarized in Table 1. Unlike previous studies, we obtained reliable and comparable results from multi-analytical approaches, allowing for a direct comparison between the two caves. Based on these results, we can ‘reconstruct’ the mural painting preparation at both caves and compare them with those in other regions along the Silk Road.

The mural paintings in Bezeklik Cave 51 and Tuyuq Cave 2 could well have been prepared in the following steps. To begin with, both caves were carved into the cliff face. The ground layer is plastered multiple times, consisting of a coarse mud layer and a fine mud layer. The coarse mud layer is applied first and contains a good amount of wheat straws and husks. A fine mud layer containing bast fibers sourced from hemp is then plastered on top of the coarse mud layer. A layer of gypsum is

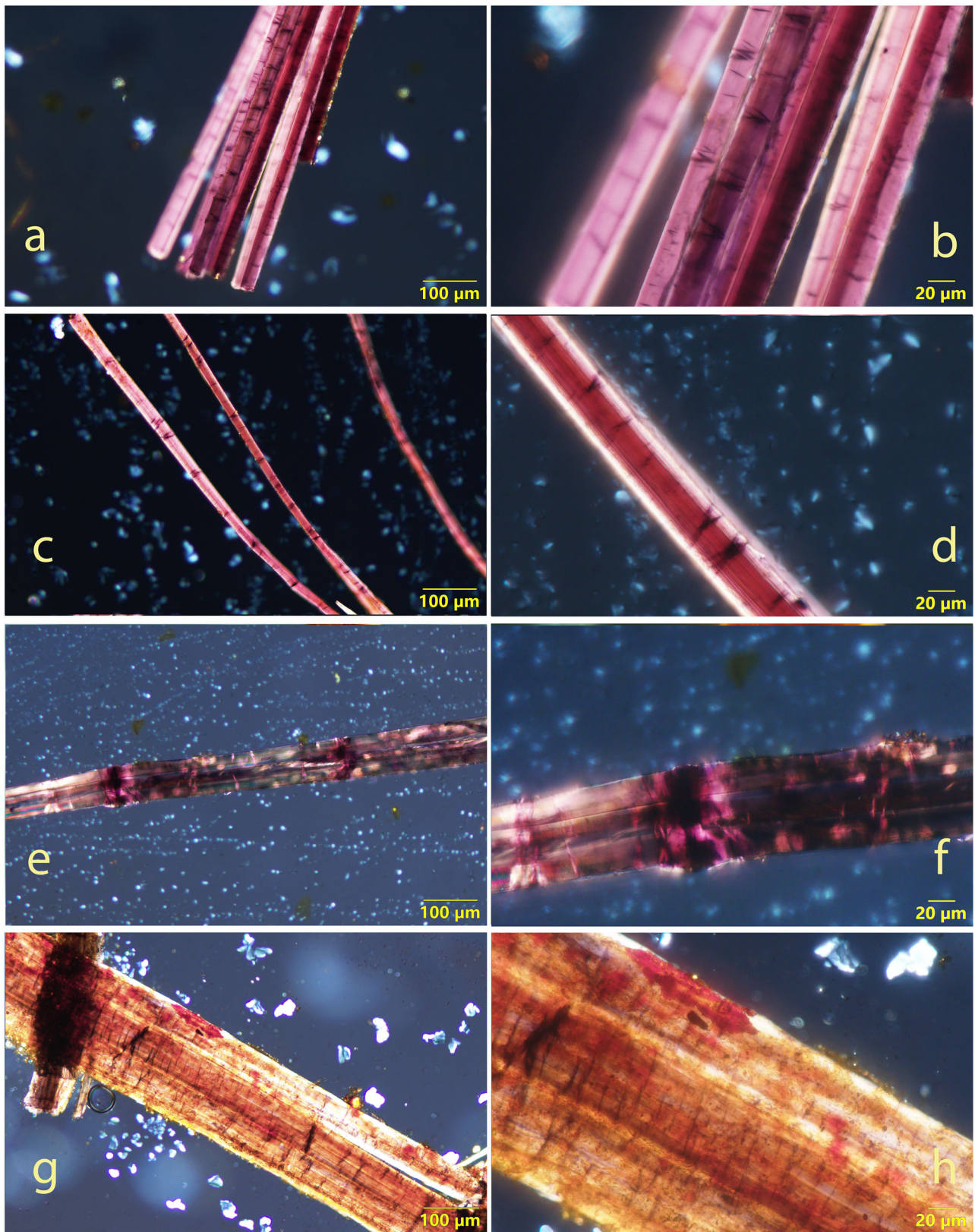


Fig. 10 | The stained color and morphology of the plant fibers that are noticed in the ground layer of mural paintings in Bezeklik Cave 51. **a–f** Short, wine-red fiber bundles identified as hemp fibers. **g, h** Yellow and partly red fiber bundles, likely unprocessed bast fibers.

brushed over the fine mud layer. Mural painters sketched and outlined against the gypsum-white background of the ground layer. Figure 14 illustrates the preparatory layers of mural paintings in Bezeklik Cave 51 and Tuyuq Cave 2.

Given the over 500-year difference between Bezeklik Cave 51 and Tuyuq Cave 2, the traditions of mural preparation seemed to have remained quite stable in Turpan and, at the same time, shared similarities with other grottoes along the Silk Road.

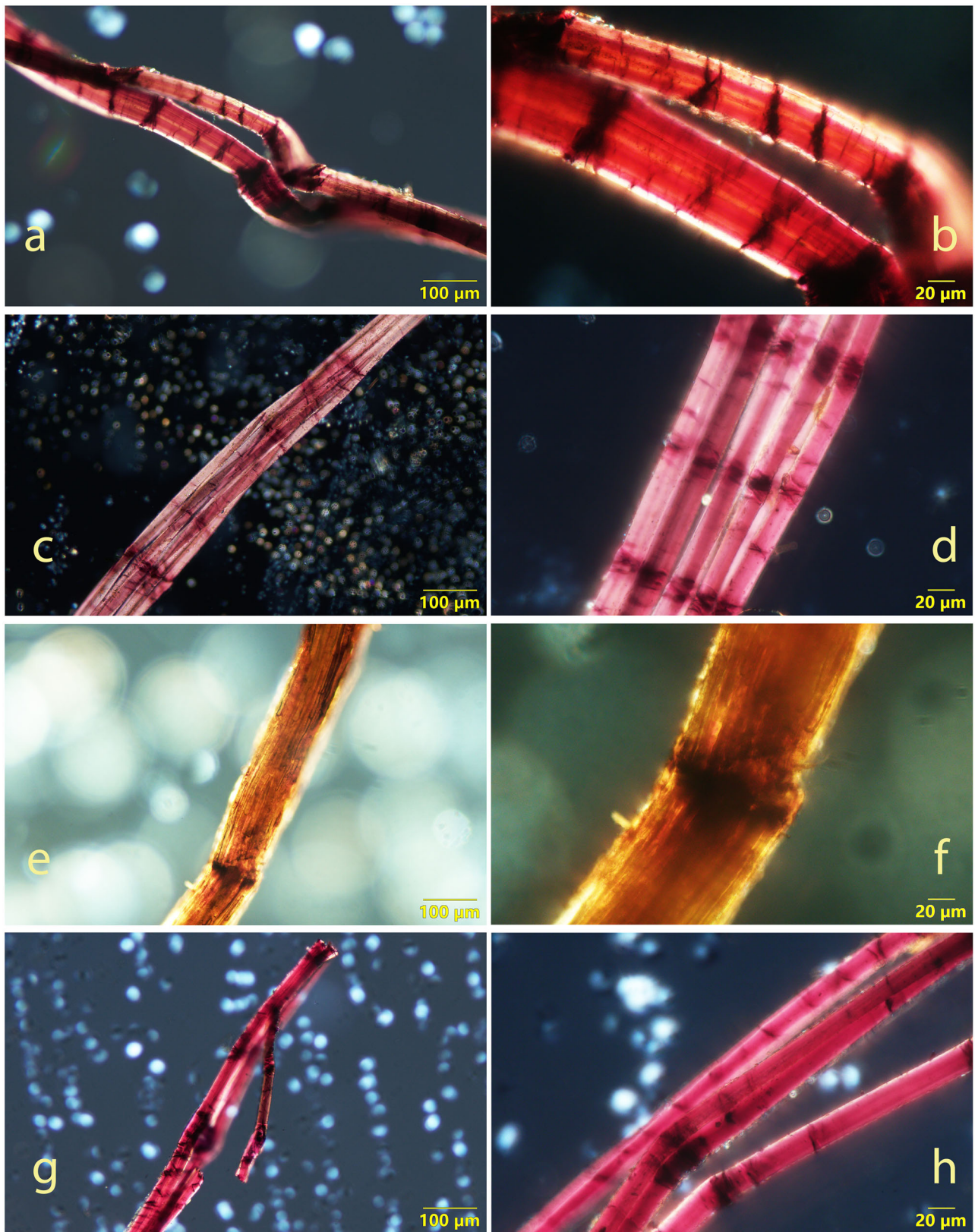


Fig. 11 | The stained color and morphology of the plant fibers that are noticed in the ground layer of mural paintings in Tuyuq Cave 2. **a–d, g, h** Wine-red fiber bundles and fibers, identified as hemp. **e, f** Fiber bundles showing a yellowish color, likely unprocessed bast fibers.

Ground layer preparation

The materials and techniques of the ground layer remain highly stable along the Silk Road. Clay, sand, and a relatively small amount of plant fibers (sometimes animal hairs as well) were mixed with water to prepare

the mud mixture. The ground layer of mural paintings in caves and grottos occasionally lacks obvious stratification¹⁷. Previous studies suggest that in Bezeklik Cave 15, the coarse mud layer contains wheat straws while the fine mud layer contains a small amount of *ma* or cotton¹¹.

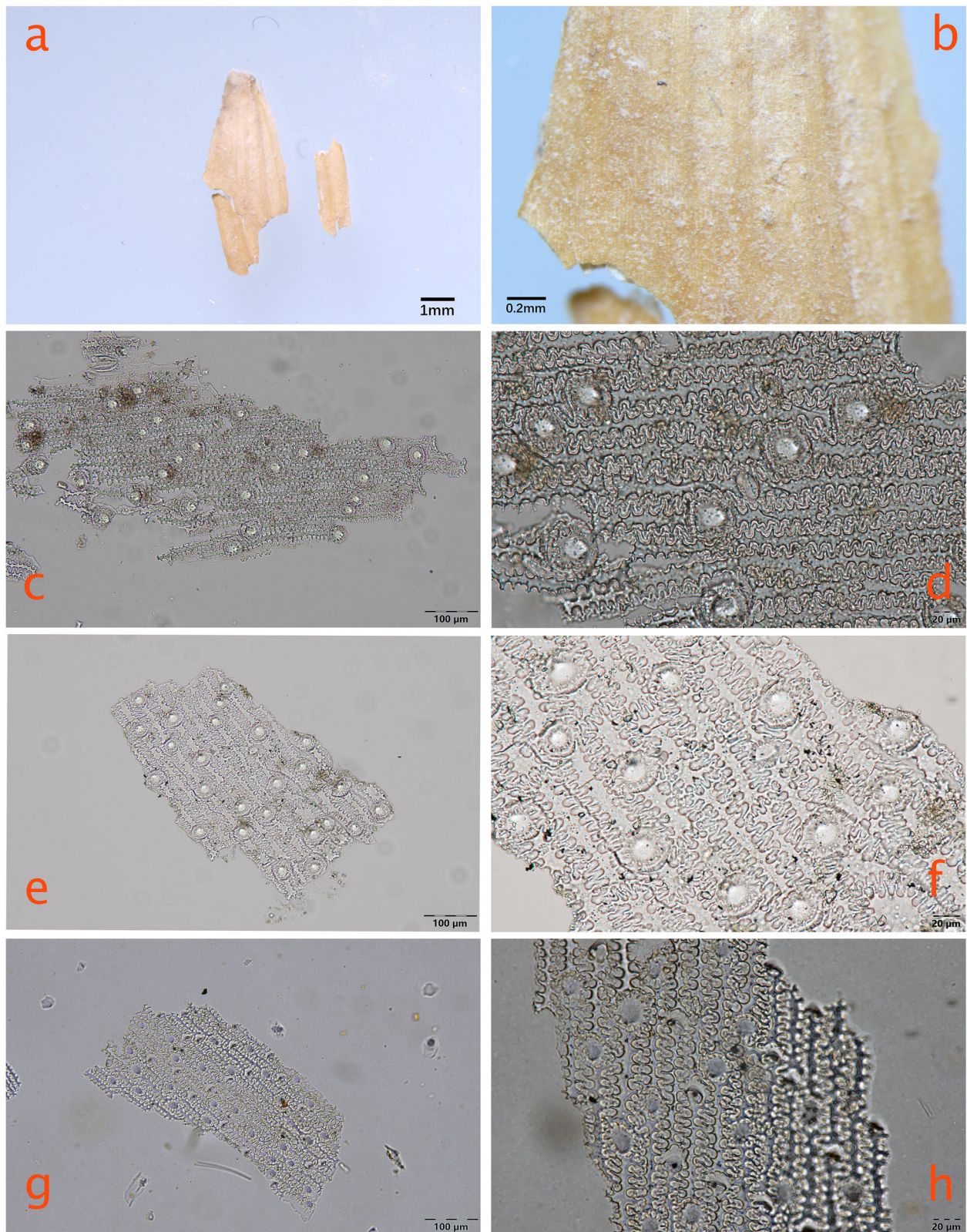


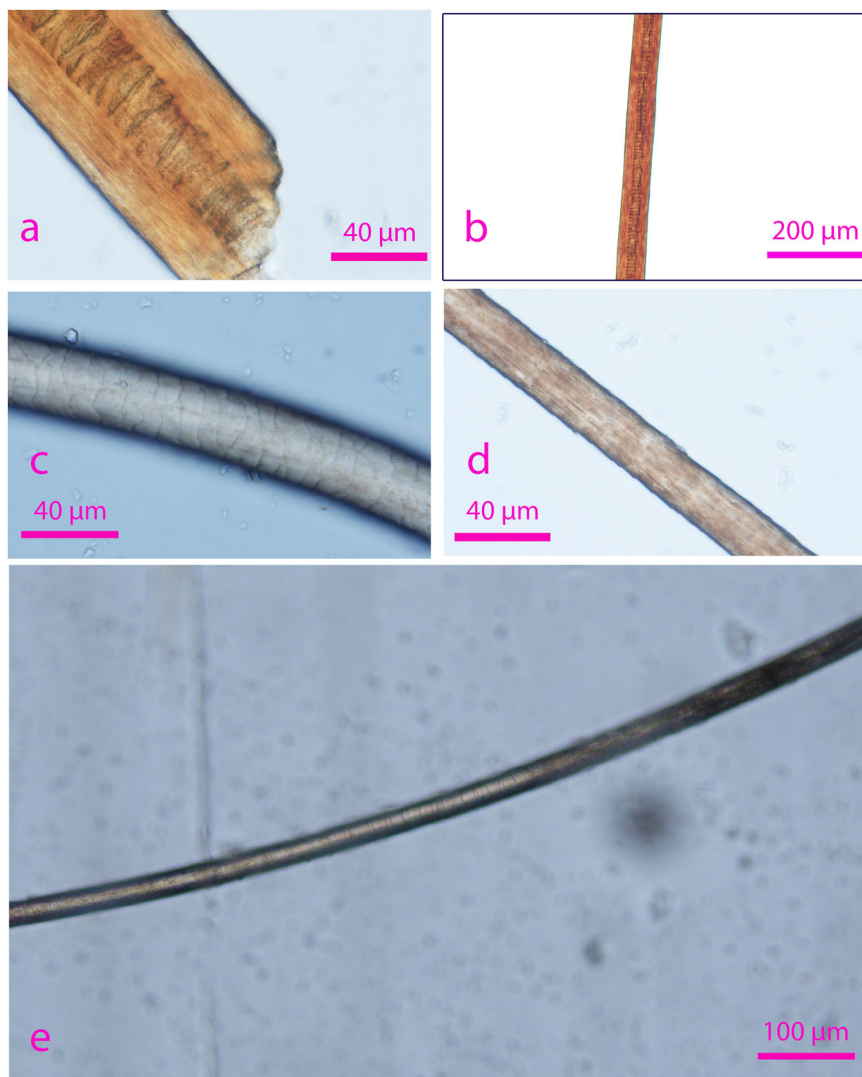
Fig. 12 | Wheat husks and phytoliths from the ground layer of mural paintings. **a, b** Husks (Bezeklik Cave 51). **c, d** Articulated elongate dendritic phytoliths and papilla phytoliths from wheat straws/husks (Bezeklik Cave 51). **e, f** Articulated

elongate dendritic phytoliths and papilla phytoliths (Tuyuq Cave 2). **g, h** Articulated elongate dendritic phytoliths and papilla phytoliths from modern wheat.

Mural paintings in Bezeklik Cave 18, 28, and 40 contain wheat straws in the coarse mud layer but fibers of *ma* in the fine mud layer¹⁷. By contrast, Tuyuq Cave 66 contains wheat straws in the coarse mud layer, without *ma* or animal fibers in the fine layer⁵. The ground layers of Bezeklik Cave

51 and Tuyuq Cave 2 contain wheat husks in the coarse mud layer and bast fibers sourced from hemp in the fine mud layer. From a technological perspective, the ground layer preparation for the two caves is largely the same as those noticed at other caves along the Silk Road (and those in

Fig. 13 | Animal fibers are noticed in the ground layer of mural paintings in Bezeklik Cave 51.
a, b Medullary cavities. c–e Hair scales.



ancient Turpan in particular), enriching utilization evidence of wheat and hemp resources in the Turpan region.

The use of pigments/dyes on mural paintings

Bezeklik Cave 51, Tuyuq Cave 2, and many other caves in Xinjiang shared the same green and black pigments as Mogao grottoes in Dunhuang of Gansu (see Supplementary Tables 4 and 5). Green colors were most often prepared from malachite and atacamite, and black colors are predominantly composed of carbon black. As noticed earlier, lead dioxide, a discoloration product of red lead, also shows a black or brown-black color. White pigments are always mineral-based, including gypsum, anhydrite, kaolin, and calcite (Supplementary Tables 4 and 5). In the early period of the Mogao grottoes, kaolin was predominant, shifting to calcite in the middle period and gypsum in the late period¹⁷. In the Xinjiang region, gypsum and anhydrite were the dominant pigments to paint white. The sources of red and blue colors are abundant, while mineral pigments are the main choices, usually iron red, cinnabar, and red lead for red, and lapis lazuli and azurite for blue (Supplementary Tables 4 and 5). Sometimes, organic dyes are also used to paint red and blue colors, such as lac dye and madder used as red colorants in the Kizil grottoes^{64–66}. Cases of indigo used in grotto murals are relatively more on spatial distribution (Supplementary Table 6), from Kucha in the west, through Gaochang (Turpan) in the middle, to Dunhuang in the east^{64,65,67,68}. Grotto murals along the Silk Road have

demonstrated continuity in the selection and use of pigments. However, differences do exist across regions, especially in white and blue pigments. Within Xinjiang, the Kucha and Gaochang grottoes predominantly use gypsum for white pigment and lapis lazuli for blue pigments. In contrast, the Mogao grottoes in Dunhuang show more pronounced temporal changes in white and blue pigments (Supplementary Table 4), possibly influenced by local mineral distribution and pigment trade along the Silk Road¹⁷.

Indigo as a blue dye

Recently, indigo blue has been discovered in the Kizil grottoes in Kucha, the Tuyuq grottoes in Turpan, and the Mogao and Tiantishan grottoes in Gansu, along the Silk Road (Supplementary Table 6), indicating a widespread spatial distribution^{64,65,67,68}. Natural indigo is a plant dye with a long history in China. The earliest recorded preparation of indigo as a blue dye can be found in *Qimin Yaoshu* (Essential Techniques for the Welfare of the People), the oldest completely surviving agricultural text of China written between 533 and 544 AD by Jia Sixie (488–556 AD) in the Northern Wei Dynasty. It states that “in the middle of the seventh month, dig a pit...cut the indigo plants and stand them upright in the pit, add water...add one *dou* and five *sheng* of lime...let it settle, drain the water, create a small pit, store the prepared indigo in the pit. When it resembles thick porridge, remove it from the jar, and it will have become indigo (七月中作坑...刈蓝倒竖于坑中,

下水...著石灰一斗五升...澄清, 泻去水, 别作小坑, 贮蓝靛放入坑中。候如强粥, 还出瓮中盛之, 蓝靛成矣”⁶⁹. The mural paintings in Tuyuq Cave 2 are radiocarbon dated to 424 to 564 AD (95.4% probability). This suggests that the application of natural indigo as a blue color in the Tuyuq grottoes was contemporaneous with, if not earlier than, the completion of *Qimin Yaoshu*. Considering the frontier location of Xinjiang, the production and distribution of natural indigo could well have been widespread at the time.

Chemically, indigo blue is supposedly less stable than mineral-based blue pigments such as azurite and lapis lazuli. Moreover, indigo blue on ancient Chinese cave murals often shows a dark blue or bluish-black color, contrasting with the high-saturation blue of mineral pigments. Beside in grotto murals, indigo was also used in ancient Chinese tomb and architectural murals, including the Tang dynasty tomb of Han Xiu in Xi'an⁷⁰, the Ming dynasty murals in the temple of the Lu family in Lutusi, Gansu⁷¹, and the mural paintings in the Yuan dynasty family tombs in the southern outskirts of Xi'an⁷². Combining these cases with use of indigo in grotto mural paintings^{64,67,68}, we noticed that sometimes indigo was used alone to paint the dark blue color, sometimes it was

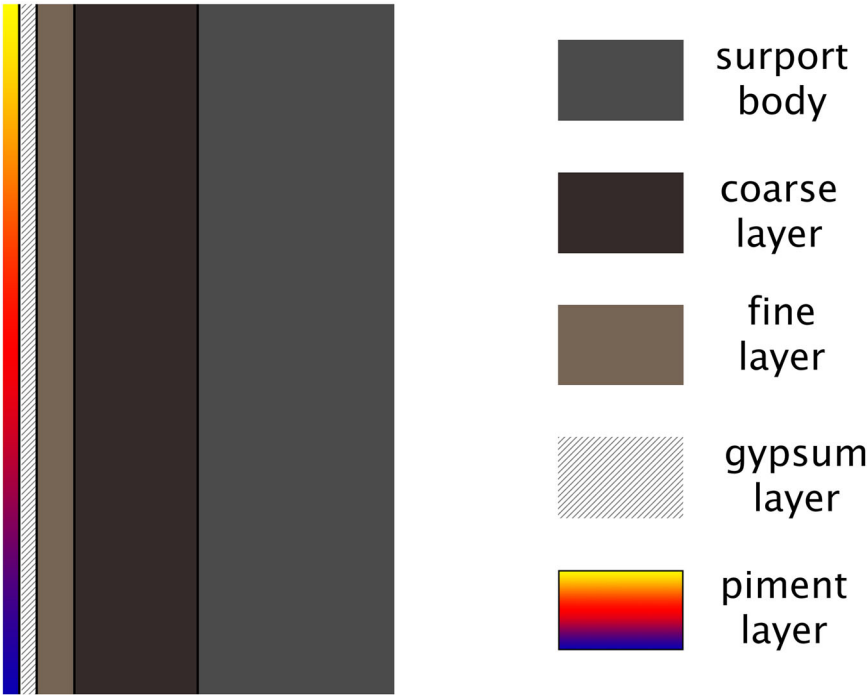
blended with mineral pigment or organic dye to express blue or other colors during the 6th to 16th century AD. We suggest that indigo, whether used independently or combined with mineral-based pigments or organic dyes, can enhance the color expression of the artwork (a similar statement has been made on Tang Dynasty painted figurines unearthed from the Astana Cemetery in Turpan⁷³. Since mineral-blue pigments have been identified in mural paintings of other Tuyuq caves^{5,6,14}, we infer that the indigo as a blue color in Tuyuq Cave 2 likely resulted from an intentional selection of raw materials for a better color effect in the mural paintings. Further analytical studies, including Raman spectroscopy and high-resolution mass spectrometry (HRMS) and a more systematic sampling of blue colors in the Tuyuq grottoes, will help address this issue.

In this study, accelerator mass spectrometry (AMS) 14C dating, Raman spectroscopy, high-resolution mass spectrometry (HRMS), Herzberg staining, and phytolith analysis were employed to characterize and identify the pigments, dyes, and fibers used in the mural paintings of Bezeklik Cave 51 and Tuyuq Cave 2 in the Turpan Basin of Xinjiang. For the first time, absolute dates were established for these caves, revealing that the mural paintings in Bezeklik Cave 51 were created between the early 11th and mid-12th century AD, while those in Tuyuq Cave 2 date to the early 5th to mid-6th century AD. Despite the over 500-year gap between the two sites, the mural preparation techniques were remarkably consistent, though variations in material selection were observed. Both caves predominantly used mineral-based pigments, with gypsum and atacamite serving as white and green colors, respectively. However, Bezeklik Cave 51 utilized red lead for red hues, whereas Tuyuq Cave 2 employed iron oxide. Additionally, the organic blue dye indigo was identified in Tuyuq Cave 2 but absent in Bezeklik Cave 51. This study provides reliable and comparable identifications of the pigments, dyes, and fibers used in the mural paintings of the two caves. Through Herzberg staining and phytolith analysis, new insights were gained into the use of plant fibers in the ground layer. Our findings establish a robust methodological framework for future research. Expanding this study to include additional caves at Bezeklik and Tuyuq will further elucidate the patterned use of materials in the Turpan Basin over time, offering deeper insights into the cultural and technological practices of the region.

Table 1 | Pigments and dye identified in Bezeklik Cave 51 and Tuyuq Cave 2 in this study

	Bezeklik Cave 51	Tuyuq Cave 2
White	Gypsum Anhydrite	Gypsum Anhydrite
Black	Carbon black Lead dioxide	–
Green	Atacamite	Atacamite
Blue (bluish-black)	–	Indigo
Red	–	Hematite
Yellow	Red lead	–
Plants in the ground layer	bast fibers (hemp) common wheat husks and straws	bast fibers (hemp) common wheat husk and straws

Fig. 14 | Illustration showing the mural painting preparation for Bezeklik Cave 51 and Tuyuq Cave 2.



Data availability

All data supporting the conclusions of this article can be obtained from the corresponding author, T.L., upon request.

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Author contributions

T.L., Y.Y. Mei, Y.Z. Chen, and X.W. conceived the study and were significant contributors to the writing of the manuscript. T.L., Y.Y. Mei, X.C. Geng, and Y.W. interpreted the Raman spectra and HRMS results. X.W., Y.Y. Mei, X.C. Geng, and Y.W. conducted the Herzberg staining and phytolith analysis. All authors reviewed and approved the final manuscript.

Competing interests

T.L., the corresponding author of this manuscript, serves as an Associate Editor of NPJ Heritage Science. However, he has not been involved in the peer review process or editorial decisions regarding this submission. The authors declare no competing interests.

Additional information

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